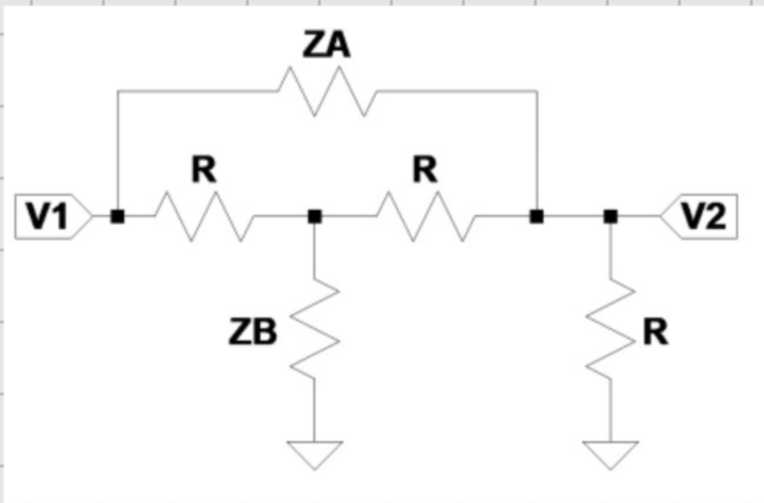
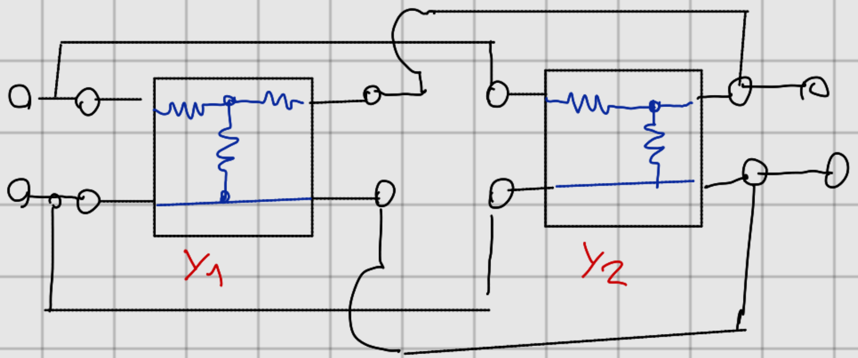
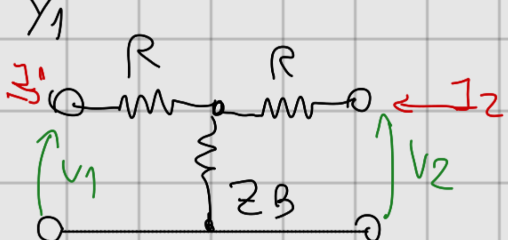


se puede dividir en 2 cuadripolos, uno de ellos tee.



como están en paralelo, trabajo con parámetros ADMITANCIA

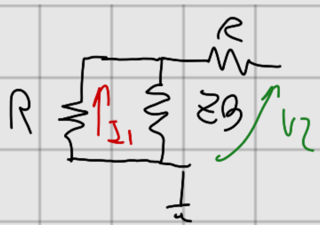


$$Y_{11} = \left.\frac{I_1}{V_1}\right|_{V_2=0} = (G+Y_B) /\!/ G$$

$$Y_{11} = \frac{\dfrac{R+ZB}{R\cdot ZB}\cdot\dfrac{1}{R}}{\dfrac{1}{R}+\dfrac{R+ZB}{R\cdot ZB}}$$

$$\boxed{Y_{11}} = \frac{\dfrac{R+ZB}{R^{\cancel{2}}\cdot \cancel{ZB}}}{\dfrac{ZB+R+ZB}{\cancel{R}\cdot\cancel{ZB}}} = \boxed{\frac{R+ZB}{(R+2ZB)\cdot R}}$$

$$Y_{12} = \left.\frac{I_1}{V_2}\right|_{V_1=0}$$



$$I_1\Big|_{V_1=0} = V_2 \cdot \frac{R//Z_B}{R//Z_B + R} \cdot \frac{1}{R} \cdot (-1)$$

$$\boxed{Y_{12}} = -\frac{1}{R} \cdot \frac{\dfrac{R \cdot Z_B}{R + Z_B}}{\dfrac{R \cdot Z_B}{R + Z_B} + R} = -\frac{1}{R} \cdot \frac{R \cdot Z_B}{2R \cdot Z_B + R^2} = \boxed{-\frac{Z_B}{2R \cdot Z_B + R^2}}$$

$$Y_{21} = \left.\frac{I_2}{V_1}\right|_{V_2=0}$$

$$I_2\Big|_{V_2=0} = V_1 \cdot \frac{R//Z_B}{R//Z_B + R} \cdot \frac{1}{R} \cdot (-1)$$ idem Y_{12}..

$$\boxed{Y_{21} = Y_{12}}$$

$$Y_{22} = \left.\frac{I_2}{V_2}\right|_{V_1=0}$$

$$Y_{22} = \left(R//Z_B + R\right)^{-1}$$

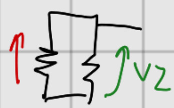
$$\boxed{Y_{22}} = \left(\frac{R \cdot Z_B}{R + Z_B} + R\right)^{-1} = \left(\frac{R \cdot Z_B + R^2 + R Z_B}{R + Z_B}\right)^{-1} = \boxed{\frac{R + Z_B}{R^2 + 2 R Z_B}}$$

$$Y_1 = \begin{pmatrix} \dfrac{R + Z_B}{R(R + 2Z_B)} & -\dfrac{Z_B}{R(R + 2Z_B)} \\[2ex] -\dfrac{Z_B}{R(R + 2Z_B)} & \dfrac{R + Z_B}{R(R + 2Z_B)} \end{pmatrix}$$

Luego, Y_2...

$$\boxed{Y_{2_{11}} = \left.\frac{I_1}{V_1}\right|_{V_2=0} = \frac{1}{Z_A}}$$

$$\boxed{Y_{2_{12}} = \left.\frac{I_1}{V_2}\right|_{V_1=0} = -\frac{1}{Z_A}}$$



$$\boxed{Y_{2_{21}} = \left.\frac{I_2}{V_1}\right|_{V_2=0} = -\frac{1}{Z_A}}$$

$$\boxed{Y_{2_{22}} = \left.\frac{I_2}{V_2}\right|_{V_1=0} = \frac{1}{Z_A} + \frac{1}{R} = \frac{Z_A + R}{Z_A \cdot R}}$$

Entonces

$$Y_2 = \begin{bmatrix} 1/Z_A & -1/Z_A \\ -1/Z_A & (Z_A+R)/Z_A \cdot R \end{bmatrix}$$

Finalmente, Y_{TOTAL}...

$$Y_{TOT} = Y_1 + Y_2$$

$$Y_{TOT} = \frac{1}{R(R+2Z_B)} \cdot \begin{bmatrix} R+Z_B & -Z_B \\ -Z_B & R+Z_B \end{bmatrix} + \frac{1}{Z_A}\begin{bmatrix} 1 & -1 \\ -1 & \frac{Z_A+R}{R} \end{bmatrix}$$

$$R \cdot Z_A + \cancel{Z_A Z_B} + Z_A R + \cancel{2 Z_A Z_B} + R^2 + R \cdot 2 Z_B$$

$$Y_{TOT} = \frac{1}{Z_A \cdot R \cdot (R+2Z_B)} \cdot \begin{pmatrix} Z_A \cdot (R+Z_B) + R(R+2Z_B) & -Z_B \cdot Z_A - R(R+2Z_B) \\ -Z_B Z_A - R(R+2Z_B) & (R+Z_B)Z_A + (Z_A+R)(R \ldots \end{pmatrix}$$

$$Y_{TOT} = \frac{1}{Z_A \cdot R \cdot (R+2Z_B)} \begin{pmatrix} R \cdot (R+2Z_B+Z_A) + Z_A Z_B & -R(R+2Z_B) - Z_A Z_B \\ -R \cdot (R+2Z_B) - Z_A Z_B & 3 \cdot Z_A Z_B + R \cdot (R+2Z_A+2Z_B) \end{pmatrix}$$

con $Z_A = \frac{R}{Z_B}$

$$Y_{TOT} = \frac{1}{\frac{R^3}{Z_B} + 2R^2} \cdot \begin{pmatrix} R \cdot (R+2Z_B+\frac{1}{Z_B}+1) & -R \cdot (R+2 \cdot Z_B+1) \\ -R(R+2 \cdot Z_B+1) & R \cdot (R+2\frac{R}{Z_B}+2Z_B+3) \end{pmatrix}$$

Para comprobar con simulaciones, asigno $R = Z_B = 1$ (y $Z_A = 1$)

$$Y_{TOT} = \frac{1}{3} \cdot \begin{pmatrix} 5 & -4 \\ -4 & 8 \end{pmatrix}$$

$$\boxed{Y_{TOT} = \begin{pmatrix} \frac{5}{3} & -\frac{4}{3} \\ -\frac{4}{3} & \frac{8}{3} \end{pmatrix}}$$